Review and refinement of the SubexponentialSAT proposals

Second round research report

27 October 2026

Abstract

A smaller unrestricted logarithmic order is not established. The user's target is stronger than the paper's exponent $O(n \log\log n/\sqrt{\log n})$, not merely a smaller constant or a new proof of that rate. The recent ProveIt arrivals preserve this order; their stronger targets remain conditional. We derive a concrete improvement on their normalized tensor interface:

$$(s+\mathcal{L})^{O(1)}2^{(84.0662714943\ldots+o(1))s/\sqrt{\lceil \log_2 s\rceil}} \quad \text{with polynomial space,}$$

with a conservative eventual coefficient 85, compared with the supplied 648.4411 certificate. The new ingredients are deterministic matching-based collision packing, a thinner precision gap, exact crossing-packet accounting, and a two-stage grouping schedule. This is a mathematical derivation relative to unchanged source identities, not an independently verified full SAT algorithm, a Lean theorem, a practical benchmark, or a coefficient directly on n. We also identify precise missing interfaces for a true order improvement and prove narrow barriers to several shortcuts. The canonical source relocation is byte-preserving and adds no new theorem.

1 What this review establishes

Fix clause width $k = 3$. Write n for the number of variables, m for the number of clauses, $L_n = \log_2 n$, and $\ell_n = \log_2 \log_2 n$; small inputs are handled separately. Polynomial factors include input reading and exact arithmetic. The imported OpenAI manuscript claims an exact-counting, deterministic polynomial-space bound

$$\mathrm{poly}(n,m)2^{O(n\ell_n/\sqrt{L_n})}, \tag{1}$$

and, at each fixed density $m \le Dn$, the stronger exponent $O_D(n/\sqrt{L_n})$ [2]. The meaningful primary target here is an unrestricted exponent $o(n\ell_n/\sqrt{L_n})$, with all algorithmic and space costs paid.

The strongest available new conclusion is a coefficient improvement. Section 4 proves the displayed 84.0663 contraction certificate on exactly the normalized interface used by the supplied 648.4411 certificate. At most s edges, maximum degree $\lceil s^{0.9}\rceil$, initial precision and truncation parameter at most s, and the source's entry-oracle allowance are essential normalizations. Replacing s by $O(n\log\log n)$ is not a license to put 84.0663 before $n\log\log n$: the reduction's hidden constants remain.

The stronger proposal already present in the precision packet would give $O(n\ell_n/L_n)$ if its structured-summation hypothesis were proved. That is asymptotically stronger than our coefficient result and remains unproved. We do not call a successful synthesis of constants a solution to the user's stronger-order request.

Trust boundary. "Derived" and "proved relative to the source" below refer to ordinary mathematical arguments conditional on the unchanged source deformation, integrality, recovery and recursive-factorization identities. Local exact checks are regression evidence, not a substitute for an asymptotic proof. No imported program, proof build or installer was run. No full SAT solver or full formal proof was constructed or executed.

2 Canonical sources and what the recent commits changed

The current frozen ProveIt snapshot is

`047f4f0426fa38d1fe02f194aa6aac254c42f643`.

The canonical project is `Computability/SubexponentialSAT`. Its research tree contains the seven supplied packets and the separate alternate manuscript. The original OpenAI source is vendored under `lib/openai-math/preprints/`.

UTC on 27 October	Commit prefix	Change
00:42:51	`a505bd768bce`	First five incoming artifacts
00:44:39	`64b3202ee84f`	Precision, parity and ETH-density packets
00:48:56	`8afdbbea48d6`	Our prior asymptotic-analysis packet
00:58:36	`2b925d8800cd`	Canonical project placement
01:03:12	`3e096089911f`	Original manuscript vendoring
01:04:24	`770edf7756d1`	Earlier canonical snapshot
01:13:52	`047f4f0426fa`	Closing snapshot and README comparison

The closing check found the Research tree exactly unchanged; only the README comparison grew. Its claim about removing a denominator should be read carefully: the linear-wire conditional interface removes a log-log numerator, not the square-root denominator. The barriers remain representation-specific. This closing README adds no theorem.

Independent byte mapping verifies all **120 filed research files**: 17 Cauchy, 15 contraction, 9 SAT-density, 16 precision, 12 parity, 15 ETH-density, 35 prior-analysis files, and one alternate PDF. Thirteen archive members were omitted as declared: five checksum ledgers and eight duplicate source files. The original paper is represented by its vendored copy. Six vendored manuscript files and the Apache license match the previously authenticated OpenAI bytes. The new README and omission index explain provenance and trust status; they do not supply a new mathematical proof. In particular the README explicitly says these materials have not been checked in ProveIt, no project Lean/Rocq development is included, and the alternate manuscript's claimed Lean development remains unpublished.

The earlier repository-wide path inventory covered 223,088 descendants, resolved all four truncated tree replies by splitting them, and recorded no query failures. It was a path inventory, not a content search of every ambiguously named file or external submodule. The final relocation check inventoried the canonical and vendor subtrees, not the entire repository again. The archive preserves compact inventories, exact file maps and hashes rather than the enormous full path listing.

Two OpenAI repository pins appear in the supplied reports:

`7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`
`42907c9e999cc00d6abe7a50bfc1023250edd1b9`.

Their four relevant TeX blobs are identical, including quantitative contraction and reduction. Thus these differing pins do not introduce a mathematical source mismatch.

3 Comparison of the proposals

The labels below are short names for the canonical Research directories; the bibliography provides immutable links. All order comparisons suppress fixed-width constants.

Packet	Derived contribution	Order status
Cauchy modes	Integral separated features with weights $1+b+2v_2(b!) = 3b+1-2s_2(b)$; leading entropy coefficient divided by $\sqrt{3}$.	Same $n\ell_n/\sqrt{L_n}$ order.
Contraction	Exact taper condition count; leading precision loss $24 \to 6$; retuned schedule; streamed finite differences. Explicit normalized certificate $18535.7538 \to 648.4411$ under its accounting.	Same general and sparse orders.
SAT density	Uniform profile $n\min\{\log(2+m/n), \ell_n\}/\sqrt{L_n}$ via density-sensitive codes and actual query sparsity.	General order unchanged; fixed density gives $n/\sqrt{L_n}$.
Precision	Exact valuation loss and minimax taper, again with leading loss 6. Compressed rank graph has high-degree group tensors.	$n\ell_n/L_n$ general and n/L_n sparse require an unproved summation hypothesis.
Parity refinement	Scalar exact parity lift; incidence degree $k(2n+a) \to k(n+a)$ at precision $n+1$; removes the truncated coefficient axis.	Same proved order. Compact factorization would give $n\ell_n^{3/2}/L_n$.
ETH density	Largely the same density profile, different conservative parameters, matched parameter coefficient improvement and exact odd-Cauchy Smith profile.	Not a second density factor to multiply into the first.
Prior analysis	Our already delivered scoped barriers and degree-sensitive compressed-kernel research gate.	$n\sqrt{\ell_n/L_n}$ remains conditional on a kernel and recursive closure.
Alternate PDF	Different 49-page manuscript attributed within the PDF to Claude / Anthropic; a growing-node crossing construction.	Same fixed-width order; formalization claimed but unavailable for audit.
This report	Matching-potential packing, thin precision gaps, crossing fraction, two-stage schedule; normalized $C < 85$ eventually.	Same order; not stronger than the conditional targets above.

3.1 The improvements that can be composed

Exact taper and precision improvements, integral factorial-Cauchy modes, and the new packing act on different parts of the contraction argument and can be composed after checking their shared precision and recursive interface. The scalar parity reduction may independently set the truncated coefficient length to one. The density refinement independently reduces the Boolean wire budget to

$$O_k(n\min\{\log(2+m/n), \log\log n\})\,.$$

Neither reduction should be multiplied blindly into a numerical coefficient on the normalized s interface. The two density packets mostly repeat the same mathematical mechanism.

The inspected density proof chain uses a lazy Margulis–Gabber–Galil walk, powering for small spectral error, exhaustive local-code search, shortening, and greedy exact-count minimization. The cost of local-code search is explicitly $2^{O(d^2)}$, not polynomial. After choosing polylogarithmic parameters, it is lower order than $2^{O(n/\sqrt{L_n})}$. High-incidence branching and degree-array encoding preserve polynomial space and yield $O(n\log(S+2))$ wires for S actual monomials per parity equation. This source audit found no fatal gap in that chain relative to the imported identities; it did not rebuild a solver or formal proof. The external expander theorem is an input, not established by tiny graph tests [10].

3.2 Why the conditional targets are not achieved

The precision packet's exact rank network has only $O(M/\sqrt{u}+r^2)$ wires in its compressed representation, but the group tensors have excessive degree. Their subexponential entry evaluators do not make them bounded-degree factors. Equality-tree splitting cannot represent an arbitrary high-arity tensor, and restoring the original child network loses the compression. Its missing summation theorem is a real algorithmic obligation.

Similarly, the parity packet's compact-factorization assumption must cover every generated group tensor, invalid-rank zeros included, with $O(W)$ total Boolean edges, maximum degree $W^{0.9}$, correct immutable creation precision, and construction/entry time $\mathrm{poly}(s)\exp(O(s/\log s + s^{0.9}))$. No such construction is supplied. The prior degree-kernel route also needs a deformation degree bound and a mode-family count measured by the compressed wire universe at every recursive child. A small root incidence degree alone does not establish these properties.

4 A combined contraction certificate below 85

4.1 Interface and scope

Let $s \geq 2$, $\lambda = \lceil \log_2 s \rceil$, and let $\mathcal{L}$ be the total input description length. The network is bipartite and may have parallel edges, has at most s Boolean edges and maximum degree $\lceil s^{0.9} \rceil$, and has entries in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad P_0, \ell \leq s.$$

Its original entry routines cost at most $\mathrm{poly}(s+\mathcal{L})2^{A_0 s^{0.9}}$ time and polynomial space, for fixed interface constants. All local tensor descriptions retain the source's immutable creation-time representatives.

Theorem 1 (Relative contraction certificate)**.** *Assuming the unchanged source deformation, integrality, recovery and recursive-factorization identities, the construction below contracts this normalized interface in polynomial space and*

$$(s+\mathcal{L})^{O(1)}2^{(C+o(1))s/\sqrt{\lambda}}, \qquad C = 84.06627149423657\ldots < 85.$$

In particular 85 *replaces* $C + o(1)$ *for all sufficiently large* s*.*

The new ingredients are near-balanced matching-potential packing, a thin additive precision gap, exact counting of crossing packets and a larger grouping alphabet at the root than in the tail. Exact taper, sharp precision and factorial-Cauchy modes come from the supplied reports and are not claimed as new here. Small or exceptional parameter choices are handled by direct summation, as in the source. The crossover can be enormous.

4.2 Deterministic near-balanced packing for multigraphs

Let an active graph have M edges and maximum degree Δ. Let $q = r^a$ be the number of labels on each side and $D_{\mathrm{arr}} = q^2$ the number of ordered label cells. Greedy edge coloring produces $K \leq 2\Delta - 1$ matchings E_j, including for multigraphs: an edge has at most $2\Delta - 2$ already colored neighbors. Write $m_j = |E_j|$.

Assign independent uniform labels in $[q]$ to vertices. For each matching and label cell γ, let $X_{j,\gamma}$ be its load. Put

$$z = 1 + \lambda^{-1}, \qquad t_j = \left\lceil (1+\lambda^{-1})m_j/D_{\mathrm{arr}} + 10\lambda^2 \right\rceil, \qquad \Phi = \sum_{j,\gamma} z^{X_{j,\gamma}-t_j}.$$

Distinct edges within a matching have independent endpoints. Independence between different matchings is neither true nor needed, since expectation is linear. Thus

$$\begin{aligned}\mathbb{E}\Phi &= \sum_{j,\gamma} z^{-t_j}(1+1/(\lambda D_{\mathrm{arr}}))^{m_j} \\ &\le K D_{\mathrm{arr}}\, e^{-5\lambda} < 1\end{aligned}$$

for all sufficiently large s in the schedules below. To see the bound per term, use

$$\log(1+1/\lambda) \ge \frac{1}{2\lambda}, \qquad (1+1/\lambda)\log(1+1/\lambda) \ge \frac{1}{\lambda}, \qquad \log(1+x) \le x.$$

The m_j terms cancel in the exponent and the additive $10\lambda^2$ leaves at most -5λ. The schedule has $D_{\mathrm{arr}} \le s$ and $K \le 4s$, so the final strict inequality holds eventually. Logarithms in this expectation estimate are natural.

Exact derandomization and bit cost. Condition on a partial vertex labeling. Different edges of one matching still depend on disjoint unassigned vertices. The probability that an edge receives cell γ is in $\{0, q^{-2}, q^{-1}, 1\}$. Therefore the conditional expectation of each summand is exactly

$$z^{-t_j} \prod_{e \in E_j} (1 + p_e/\lambda).$$

At each vertex, try all q possible labels and choose one minimizing this rational number summed over all terms. The average over labels is the current conditional expectation, so the invariant $\mathbb{E}[\Phi \mid \text{assigned labels}] < 1$ persists. At completion $X_{j,\gamma} < t_j$ for every cell and matching, since otherwise one summand would be at least one. Hence, using $\lceil x \rceil - 1 < x$,

$$X_\gamma < (1+\lambda^{-1})M/D_{\mathrm{arr}} + 10K\lambda^2 \le (1+\lambda^{-1})M/D_{\mathrm{arr}} + 20\Delta\lambda^2. \tag{2}$$

There are polynomially many vertices, tried labels, cells and edges in $s + \mathcal{L}$. Every exponent t_j is $O(s+\lambda^2)$, the rational bases have polynomial-size integer numerators and denominators, and products involve at most polynomially many factors. Exact rational products, sums and comparisons therefore have polynomial bit length and polynomial time/space cost. This deliberately simple implementation is not claimed practical.

Pad every cell to the same multiple H_{pad} of packet length $L = 2^a$. The total padded size is

$$B = H_{\mathrm{pad}} D_{\mathrm{arr}} \le (1+\lambda^{-1})M + 20 D_{\mathrm{arr}}\Delta\lambda^2 + L D_{\mathrm{arr}}. \tag{3}$$

Consequently $B = (1+o(1))M$ whenever $D_{\mathrm{arr}}\Delta\,\mathrm{polylog}(s) = o(M)$ and $LD_{\mathrm{arr}} = o(M)$. This replaces the source's sufficient $D_{\mathrm{arr}}^2\Delta = o(M)$ condition while leaving its subsequent packet arrangement intact. There are B/L packets. Each coordinate-symbol class has at most B/r real ports on either side, so a child has at most

$$2B/r = (2/r)(1+o(1))M$$

edges. New packet vertices have degree at most $2L$; retained vertices keep their old degrees.

A sharp scale for this specific packing architecture. Take t disjoint pairs, with Δ parallel edges on each pair, so $M = t\Delta$. Every nonzero cell load is a multiple of Δ and at least one is nonzero. Equal padding therefore forces $B \ge \Delta D_{\mathrm{arr}}$. Near-unit total padding is impossible if $\Delta D_{\mathrm{arr}}/M$ diverges. This is only a limitation of equal-padded collision packing, not a lower bound for contraction or SAT.

Count crossing packets rather than all packets. At any fixed coordinate, exactly the fraction $1-1/r$ of ordered padded label pairs have unequal symbols. Every packet fixes that pair. Thus exactly $(1-1/r)B/L$ packets can have crossing modes there. At retained valuation i, put $D_t = 2^{a-i}$. The number c of column-tile families satisfies

$$c \le 2(1-1/r)B/D_t = (2(1-1/r)+o(1))M/D_t. \tag{4}$$

Empty or purely local tiles only reduce this number.

4.3 Thin precision gaps and exact taper

Set $g(\lambda) = (\lceil \log_2 \lambda \rceil)^2$. For positive integer taper lengths u_d, use

$$a_d = \sum_{j=0}^{d} u_j + (d+1)g(\lambda), \qquad a_{-1} = 0. \tag{5}$$

The source requires the cumulative packet exponent to dominate inherited precision; it does not require literally doubling every taper increment. At a retained $i < u_d$,

$$D_t = 2^{a_d - i} \ge 2^{a_{d-1}+g+1}.$$

Use the complete linear taper, with $a = a_d$, $u = u_d$,

$$h = \left\lceil \frac{M(2L-1)+1}{L(2u-2+2^{1-u})} \right\rceil, \qquad q_i = h(u-i) \quad (0 \le i < u), \tag{6}$$

and $q_i = 0$ thereafter. The exact condition count is

$$\sum_{i=0}^{u-1} \frac{L}{2^i}(u-i) = L(2u-2+2^{1-u}),$$

so the total multiplicity is greater than the deformation degree $M(2L-1)$. The supplied sharp precision identity is

$$p_t = p + B_i + (q_i - 1)(a+1-i), \qquad B_i = D_t \sum_{j=1}^{a-i} j2^{-j}(q_i - q_{i+j}) \le 6hD_t. \tag{7}$$

The taper is nonincreasing, its precision increments are positive, and the immutable creation-precision lifting convention is unchanged.

Along a recursion path, precision increments are at most $s\lambda^{O(1)}2^{a_d}$. Inherited precision is therefore at most $s\lambda^{O(1)}2^{a_{d-1}}$. With leaf threshold $H = \lceil s/\lambda \rceil$, active calls have $M > s/\lambda$ and $u = O(\lambda)$, so inherited precision divided by MD_t/u is at most $\lambda^{O(1)}2^{-g} = o(1)$. The last evaluation-loss term in (7) is negligible for the same reason. If u grows as a positive power of λ uniformly at active depths, then $h = (1+o(1))M/u$ and

$$p_t \le (6+o(1))MD_t/u. \tag{8}$$

The depth proof below supplies exactly that uniform growth. Over $O(\log \lambda)$ levels, the total gap consumes $O((\log \lambda)^3) = o(\lambda)$ label coordinates. No extra root precision is assumed.

4.4 Factorial modes and the complete arithmetic interface

Use the supplied exact Cauchy features, whose guaranteed weights are

$$w_b = 1 + b + 2v_2(b!) = 3b + 1 - 2s_2(b), \qquad 0 \le b < D_t/2,$$

where $s_2(b)$ is binary popcount. The actual row parameters satisfy the required prime-subring congruence at every interpolation value: $z_x/2$ differs from a rational 2-integral quantity $2^i/d_0$ by a multiple of 2^{a+1-i}. Since $b < D_t/2 = 2^{a-i-1}$, the factorial-divisibility argument applies. There is no division by a factorial in a residue ring. Binomial features are computed as ordinary integers; $b!$ is multiplied into the target coefficient. Introduced denominators are units. Changing r, the taper or the packing does not change these local facts.

For c families and precision p, the generating-product estimate is

$$\log N(p, c) \le \left(\frac{\pi\sqrt{2}}{3} + o(1) \right) \sqrt{cp}.$$

Apply monotonicity using uniform upper envelopes

$$P = (6 + o(1))MD_t/u, \qquad C_{\text{fam}} = (2(1 - 1/r) + o(1))M/D_t,$$

not an assumption that actual populated families or actual precision attain these envelopes. Their ratio has scale D_t^2/u, and $D_t \ge 2^g$, so the small-saddle regime and its errors hold uniformly. Finite tile truncation only decreases the count. Combining (4) and (8) gives the local natural-log coefficient

$$\log N \le \left[2\pi\sqrt{\tfrac{2}{3}(1 - 1/r)} + o(1) \right] M/\sqrt{u}. \tag{9}$$

The exact local kernel, creation-time representatives and recursive child definitions are unchanged. The original source's packet-entry enumeration costs $\exp(O(L \log s))$, which is already lower order for the schedule below. The supplied streamed finite-difference replacement can reduce this to $\text{poly}(s + \mathcal{L})2^L$ using at most L extra internal precision bits, but it is optional and not needed for this theorem. If used, the numerator must be computed modulo $2^{p_{\text{creation}} + v_2(\prod \beta!)}$ before dividing canonical coordinates by the explicit power of two. Temporary precision must not alter tensor creation precision.

The extension field still has degree 4λ: deformation degrees $M(2L - 1)$ are less than s^4 eventually. All precisions, field coordinates, rational-recovery operands and stored descriptions have polynomial bit length. The initial entry allowance $2^{A_0 s^{0.9}}$ is lower order than the target exponent. Old entry routines are not recursively composed; descriptions remain immutable. The bound permits $\ell \le s$; the scalar parity specialization $\ell = 1$ is compatible but unnecessary here.

4.5 An explicit two-stage grouping schedule

Choose exact rationals

$$r_0 = 37, \quad u_0 = \left\lfloor \frac{10}{1043}\lambda \right\rfloor; \qquad r_d = 8, \quad u_d = \left\lfloor \frac{\lambda}{236(5/2)^{d-1}} \right\rfloor \quad (d \ge 1). \tag{10}$$

Use (5) and $H = \lceil s/\lambda \rceil$. First consider all candidate depths $d \le N = \lceil \log_3 \lambda \rceil + 3$, without assuming this is the actual depth. For these candidates the total gaps are $O((\log \lambda)^3) = o(\lambda)$, and

$$a_0/\lambda \le 10/1043 + o(1), \qquad a_d/\lambda \le A + o(1), \quad A = \frac{10}{1043} + \frac{5}{708} = \frac{12295}{738444}.$$

The packing exponents are strictly less than one:

$$0.9 + 2(10/1043)\log_2 37 = 0.999893640759903\ldots < 1, \tag{11}$$

$$0.9 + 2A\log_2 8 = 0.9 + 6A = \frac{307654}{307685} < 1. \tag{12}$$

The first follows from the exact integer inequality $37^{200} < 2^{1043}$; the second is rational. Thus $D_{\mathrm{arr}}\Delta\lambda^3/s = o(1)$ uniformly in the candidate range. Also $LD_{\mathrm{arr}}\lambda/s = o(1)$: its root exponent is $(10/1043)(1 + 2\log_2 37)$ and its tail exponent is $7A$, both below one. Finally $2L \le \lceil s^{0.9} \rceil$ eventually, preserving the degree interface.

Noncircular depth bootstrap. For every candidate active tail call, these estimates give $B/M \le 4/3$ eventually, so each child has at most $M/3$ edges. Starting from $M \le s$, an active call cannot reach depth N; it would already have at most H edges. Hence every real active call lies in the candidate range, closing the induction. This coarse bound ensures u_d grows as a positive power of λ, since $5/2 < 3$. More sharply, (3) gives $B/M = 1 + O(1/\lambda) + o(1/\lambda)$ uniformly. Over $O(\log\lambda)$ levels this accumulates to $1 + o(1)$, yielding for $d \ge 1$

$$M_d \le (1 + o(1))s\frac{2}{37}\,4^{-(d-1)}. \tag{13}$$

An active tail depth is at most $\log_4\lambda + O(1)$, and therefore

$$u_d = \Omega(\lambda^{\eta+o(1)}), \qquad \eta = 1 - \frac{\log(5/2)}{\log 4} = 0.33903595\ldots > 0.$$

This justifies uniform floor errors, exact-taper asymptotics and the inherited-precision estimates. The finite construction can check all guards and use direct summation when they fail.

4.6 The coefficient and total resources

Write

$$K(r) = \frac{2\pi}{\ln 2}\sqrt{\frac{2}{3}(1 - 1/r)}.$$

Convert (9) to base two and sum along (13). The geometric ratio in the tail is $\sqrt{5/2}/4 < 1$. Thus

$$\begin{aligned} C &= \frac{K(37)}{\sqrt{10/1043}} + \frac{(2/37)K(8)}{\sqrt{1/236}\,(1 - \sqrt{5/2}/4)} \\ &= 84.06627149423657\ldots < 85. \end{aligned} \tag{14}$$

An elementary rational upper certificate uses

$$\pi < 22/7, \quad \ln 2 > 69/100, \quad \sqrt{12516/185} < 8226/1000,$$

$$\sqrt{413/3} < 11734/1000, \qquad \sqrt{5/2} < 1582/1000.$$

These imply

$$C < \frac{9128012476}{108030195} = 84.49501064\ldots < 85.$$

The logarithm lower bound also follows by taking the first three positive terms of $\ln 2 = 2\sum_{j\ge 0}(1/3)^{2j+1}/(2j+1)$. An independent checker uses slightly different rational envelopes and obtains another strict upper bound $1659638632/19727253 < 85$.

Polynomial branching overhead over $O(\log \lambda)$ levels contributes $O(\lambda \log \lambda)$ to the exponent, while leaf work contributes $O(s/\lambda)$. Both are lower order. So is original packet enumeration, since $L = s^{A+o(1)}$ with $A < 0.02 < 0.9$; in particular this entry cost also fits the original $2^{O(s^{0.9})}$ allowance. Depth-first accumulation stores only polynomially many bits. Branching depends on s, not on $\mathcal{L}$, so the dependence on description length remains a fixed polynomial. This completes the relative proof of Theorem 1.

4.7 What the numerical comparison does and does not mean

Declared normalized construction	Leading certificate
Supplied contraction schedule	648.4410666487
That schedule plus factorial weights alone	374.3776 . . .
Matching packing with one fixed alphabet	93.8049738133
This two-stage schedule	84.0662714942
Conservative eventual certificate here	85

These are comparable certificates, not measured speeds or lower bounds on earlier algorithms. The 648.4411 construction uses geometric modes, $r = 256$, threefold padding and doubled cumulative taper; the naive factorial substitution divides its leading coefficient by $\sqrt{3}$ after compatibility checks. A single-stage matching construction with $r = 46$, taper decay 8 and initial taper fraction $10/1263$ yields the intermediate value shown. The two-stage schedule is not claimed globally optimal.

All these bounds still have denominator $\sqrt{\log s}$. The unrestricted SAT wire budget remains $O_k(n \log \log n)$ and the fixed-density budget $O_{k,D}(n)$. Optimal finite taper profiles are compatible but have the same leading six-bit loss; they do not supply a new logarithmic power. A true order improvement needs a different summation or representation idea, not merely further tuning of these constants.

5 What the stronger order problem still requires

There are three distinct constructive routes, not three established ingredients.

1. **A degree-sensitive compressed deformation.** Produce a recursively closed kernel with global degree K and mode multiplicity $O(M/D_t)$ where M is the compressed wire count. An exterior-power identity with degree $K = O(n)$ is insufficient if its modes still count the uncompressed incidence universe.
2. **Structured mode aggregation.** Factor or sum the genuine high-arity rank tensors within the precision packet's target cost, preserving invalid-index zeros, bounded degree, polynomial descriptions, exact entry costs and polynomial space.
3. **A different effective crossing algebra.** In the alternate manuscript's framework, construct a graded family with entropy exponent below $1/2$ while preserving exact crossing reconstruction, both factorizations, locality or a valid replacement, and effective integer costs.

Any proposed improvement must retain deterministic exact arithmetic, polynomial-sized coordinate precision, a fixed polynomial dependence on description length, and immutable creation-time representatives. Reducing a leaf cutoff does not remove the root entropy. Exponentially many simultaneously stored tensor entries are not compatible with polynomial space. These are mathematical open interfaces, not denied permissions or missing tool access.

6 What the actual Cauchy rows can and cannot improve

The density-refinement intake already proves the Smith profile below (its Section 9.4, equation labelled `eq:smith`). We do not claim that theorem as new. The following connects it explicitly to actual packet slots and states the resulting exact approximation obstruction. All rank statements concern integral separated approximations to this local kernel, not the complexity of SAT.

Actual packet subblock. At a retained node $t = 2^i$, choose the allowed interpolation value $\zeta = 1$, so $T = 2^i + 2^{a+1}$. In a packet whose two labels differ in coordinate i, take slots $x = 2^{i+1}k$ and $y = 2^{i+1}j$, with $0 \le k, j < J = 2^{a-i-1}$. Both slots have bit i equal to zero, so these are crossing ports. Set

$$\delta = 1 + 2^{a+1-i}, \qquad A = 2^{-i}\mathcal{C}(T)/\mathcal{C}'(0).$$

Then their crossing block is exactly

$$W_{2^{i+1}k,2^{i+1}j}(T) = AK_{kj}, \qquad K_{kj} = \frac{1}{\delta + 2(k-j)}.$$

All denominators of K are units. Moreover $v_2(A) = a + 1 - i$: in the product for $\mathcal{C}(t + 2^{a+1})$, the factor belonging to t has valuation $a + 1$, every other factor has its unshifted valuation, and the product of these other valuations equals that of $\mathcal{C}'(0)$ by the complete residue grid. The statement uses a filled, permissible packet, not a promise that all inputs contain one. Such a block is enough to test claims that hold uniformly for all permitted local kernels.

Exact invariant factors. For the $J \times J$ matrix K, the Smith valuations are

$$e_b = 2b + 2v_2(b!) = 4b - 2s_2(b), \qquad 0 \le b < J.$$

Indeed, an q-minor on distinct integer row and column indices has valuation

$$q(q-1) + v_2 \prod_{k<l} (x_l - x_k) + v_2 \prod_{k<l} (y_l - y_k).$$

The Vandermonde on any q integers is divisible by $\prod_{b<q} b!$, as follows by factoring $b!$ from column b of the binomial-polynomial evaluation determinant. Consecutive indices attain equality. Thus the qth determinantal ideal has valuation $q(q-1) + 2\sum_{b<q} v_2(b!)$; successive differences give e_b. The same proof applies over the unramified extension used by the contraction algorithm.

Exact modular approximation rank. For any integer $h > 0$, the least inner dimension r in an integral factorization $K \equiv UV \pmod{2^h}$ is

$$r_h = \#\{b < J : e_b < h\}.$$

To prove the lower bound, apply the integral invertible Smith row and column operations. If a rank-at-most-r matrix approximated the diagonal Smith matrix to precision $h > e_r$, consider its first $(r+1) \times (r+1)$ principal block. Its determinant is zero. But the unperturbed determinant has valuation $\sum_{b \le r} e_b$; every term involving $k \ge 1$ error entries has valuation at least

$$kh + \sum_{b=0}^{r-k} e_b > \sum_{b=0}^{r} e_b.$$

Such higher-valuation terms cannot cancel the unperturbed determinant. This is a contradiction. Conversely, retain exactly the Smith diagonal entries below precision h and discard the others, then undo the integral changes of basis. For the actual unnormalized block, replace h by $P-(a+1-i)$ when approximation precision is $P > a+1-i$.

Consequently, a uniform integral separated expansion whose mode coefficients have nondecreasing guaranteed valuations w_b cannot have $w_r > e_r$ on the normalized block: retaining its first r terms would contradict the rank bound. In particular, superlinear growth $w_b/b \to \infty$ is impossible by a mere integral separated-basis change on this local family. This leaves open nonseparated summation, correlated cancellation, new deformations, or a different recursive representation.

Why slope three does not transfer automatically. The factorial-Cauchy intake proves weights asymptotic to $3b$ using an enlarged auxiliary row domain. It is incorrect to assert that the actual reciprocal rows contain every even residue. For example, $t=1$, $i=0$, and tile $y_0=0$ give crossing rows $x=2k$ and $z_x = 2/(1+2k+2^{a+1}) \equiv 2 \pmod 4$. For $i=1$, $t=2$, $y_0=0$, and $a \geq 3$, the possible rows are either $z_x \equiv 2 \pmod 4$ or $z_x \equiv 12 \pmod{16}$; residue $4 \pmod{16}$ is absent. These are congruence exclusions even at arbitrarily large packet size. Thus the exact $4b + O(\log b)$ subblock obstruction is valid, whereas an optimal $3b + O(\log b)$ obstruction on all actual rows needs a separate proof. Splitting into additional row residue classes can also increase mode-family multiplicity, so a locally larger slope alone is not an end-to-end speedup.

7 The separate alternate manuscript

The 49-page *A $2^{o(n)}$ Time Algorithm for 3-SAT* is a different document from the 67-page OpenAI paper. Its printed attribution is Claude / Anthropic; this is not independent authentication of organizational authorship. Its SHA-256 is

`f0af035b6c87d04d3fcf07dae040df98029f4990c6493e3e9ab64c4ac714c0eb`.

Its strongest printed statement is a deterministic exact #k-SAT bound

$$2^{O\left(\min\{k^2 n \log\log n/\sqrt{\log n},\, km/\sqrt{\log m}\}\right)}$$

with an absolute hidden constant. For fixed $k=3$ this has the same unrestricted and fixed-density orders as (1). Unused declared variables are accounted for by multiplying the count by 2^{N-n} after normalization.

The main body combines balanced networks and signed crossing decompositions. The row-or-column choice is global for a term, not independent at each rectangle. Appendix A states the uniform effective-piece contract and derives a logarithmic denominator power $\log_2(1/\rho)$. Appendix B's partition/Fock construction gives $\rho = 1/\sqrt{2}$, hence power $1/2$. Appendix D improves the unrestricted n dependence using sparse parity checks and modular truncation. This is substantive mathematical content, but this bounded audit did not independently reconstruct every coefficient identity or external sparsification estimate.

7.1 Formalization claims are not supplied proof evidence

Appendix E claims that both its strongest `MainTheorem` and `ETHIsFalse` were proved in Lean 4.33.1 over a pinned Mathlib, with 1,881 files and several checker paths. However, the printed listing contains definitions only, no theorem proof bodies, and the full-proof URL is literally a placeholder. The

canonical project README also states that the claimed development is unpublished. The right conclusion is *unverified formalization claim*, not "verified strongest theorem" and not "only the weaker theorem was claimed". No such build was performed here.

Outstanding source-audit obligations include the Appendix B Laurent-series locality identity, Appendix C's explicitly labeled sparsification proof sketch, precise external Calabro thresholds used in Appendix D, and the correspondence between the paper's encoding/model and the claimed formal machine. The manuscript itself notes that some external thresholds were assumed rather than checked against the dissertation. No specific contradiction was found in this bounded review; that is not a certificate of correctness.

Its own stated bound-level crossover estimates are astronomical: examples include $n > 10^{10^8}$ for the unrestricted result and beyond 10^{10000} for a fixed-density example. These are not measured runtimes. Our new normalized coefficient does not make those examples practical, and does not automatically transfer to this different construction.

7.2 A genuine entropy lever and a restricted obstruction

Appendix H observes that if the full effective construction worked with $\log\dim V_E \sim CE^\alpha$, its concavity argument would yield

$$\rho = 2^{\alpha-1}, \qquad \log_2(1/\rho) = 1-\alpha.$$

Thus $\alpha < 1/2$ could improve the actual denominator power, if every required identity and cost bound survived. The manuscript does not construct such a replacement. The following new elementary deductions show why simply thinning the existing space is not enough.

Lemma 1 (Partition states forced by the same Heisenberg relations)**.** *Let $W = \bigoplus_E W_E$ be a graded vector space over a characteristic-zero field, with $0 \neq v \in W_0$. For every positive integer m, suppose commuting creation operators a_{-m} have degree m, commuting annihilators a_m have degree $-m$, $[a_m, a_{-j}] = m\delta_{mj}I$, and $a_m v = 0$. Then $\dim W_E \geq p(E)$, the ordinary partition number.*

Proof. For each finite-support vector $\beta \geq 0$ with $\sum_m m\beta_m = E$, set $v_\beta = \prod_m a_{-m}^{\beta_m} v$. In a linear relation among these vectors, apply $A_\beta = \prod_m a_m^{\beta_m}$. If $\gamma \neq \beta$ has the same weighted sum, some coordinate satisfies $\beta_m > \gamma_m$; commuting annihilators rightward then kills v_γ. The remaining term is

$$A_\beta v_\beta = \left(\prod_m m^{\beta_m}\beta_m!\right) v \neq 0.$$

Thus the coefficient of v_β is zero. This holds for all β, proving independence. □

An elementary bound suffices: choose r with $\sum_{j=2}^{r} j \leq E$. Every subset of $\{2,\dots,r\}$, padded to energy E with ones, gives a distinct partition. Hence $p(E) \geq 2^{r-1}$ with $r = \Theta(\sqrt{E})$, and $\log\dim W_E = \Omega(\sqrt{E})$. A nonzero quotient preserving all these exact relations and the vacuum cannot achieve $\alpha < 1/2$. This is not a lower bound for SAT, tensor contraction, all graded algebras, or all crossing decompositions.

Lemma 2 (Naive infinite thinning loses ordinary finite-order locality)**.** *Retain only modes in a set $S \subseteq \mathbb{Z}_{>0}$ in the standard Heisenberg current. If S and its positive complement are both infinite, no finite H makes its scalar commutator vanish after multiplication by $(z-w)^H$.*

Proof. Its coefficient sequence is $c_j = j\mathbf{1}_{|j|\in S}$, with $c_0 = 0$. Multiplication by $z-w$ takes a forward difference after reindexing, so locality would give $\Delta^H c_j = 0$ for every integer j. In characteristic

zero this is a polynomial sequence of degree at most $H-1$. Infinitely many missing positive modes force infinitely many roots, so the polynomial vanishes identically; retained positive modes contradict this. □

Periodic thinning may yield other diagonals of locality, but not the ordinary locality used by the source. Nonlinear regrading, for example $\deg t_m = m^2$, also breaks the printed homogeneous-shift rule: the degree of a product is no longer determined by its linear mode-index sum. Linear rescaling preserves the square-root entropy power. A new construction could escape these assumptions, but must reprove exact reconstruction, both integer factorizations, polynomially many pieces, polynomial bit lengths, and effective $2^{O(\lambda)}$ evaluation. A smaller Hilbert series alone is not an algorithm.

8 Proof status and verification evidence

Claim	Status in this report	Evidence limit
Canonical relocation and vendor identity	Exact byte checks pass	Provenance, not theorem verification.
Source ETH and quantitative contraction claims	Imported hypotheses	No full independent proof or solver build.
Supplied local taper, precision, factorial and density derivations	Source-reviewed relative results	Imported identities and external theorems remain inputs.
Matching-potential packing and bit complexity	New mathematical argument	Exact finite cases are regression checks only.
Combined two-stage coefficient below 85	New relative derivation	No coefficient on SAT variable count; eventual, not practical.
Actual Cauchy rank gate	New actual-domain connection to prior Smith theorem	Restricted integral separated kernel family only.
Heisenberg and thinning barriers	New elementary deductions	Same algebra/current only; not SAT lower bounds.
Stronger unrestricted logarithmic order	Not established	Constructive interfaces remain missing.
Alternate strongest Lean formalization	Claimed in PDF, unaudited	Public proof bodies and checker artifacts unavailable.

Independent review cleared the two-stage derivation at the analytic-refinement level, conditional on the unchanged source identities. It found no packing, factorial-denominator, creation-precision or variable-alphabet obstruction. This is not an independent validation of the entire manuscript.

The bundle includes newly authored standard-library regression scripts. They check exact matching conditional expectations on multigraphs, finite taper identities, rational schedule inequalities, and the actual Cauchy minors and reciprocal-row congruences. The author checker records 608 taper cases, 42 multigraph packing cases and 240 exact conditional means. The independent checker records 17 conditional-expectation cases across 279 exhaustive completions, 12 packet-count fixtures, 2,480 extension-field factorial identities and valuations, and 30 Hermite schedules with 432 rational multiplier checks. The frontier checker records 3,804 exact Cauchy minor checks across 18 matrices, plus actual-row residue tests. These finite ranges do not establish the asymptotic theorem. The rational certificate for $C < 85$ does not rely on floating-point output. Normal and optimized Python runs must execute the same explicit checks; an `assert`-only checker would not satisfy this requirement. The release ledger

records check counts, hashes, review scope and negative controls, separate from mathematical proof claims.

The archive verifier checks safe unique paths, the complete member inventory, lengths and SHA-256 hashes, plus selected immutable source Git blobs. Corruption, missing members, extra members, duplicate paths and traversal attempts must be rejected. An isolated replay rebuilds only this authored report, compares extracted text and every page's rendered pixels, and reruns our local checks in normal and `-O` modes. Imported TeX and scripts are data; they are not compiled or executed. The manifest proves internal consistency, not authorship: the separately communicated whole-archive hash is the release identity.

9 Recommended next research steps

1. **For a genuine order improvement, prioritize structured summation.** Try to factor the actual generated rank tensors rather than a generic high-arity tensor. A candidate must state exact construction, Boolean edge count, degree, entry bit-cost, precision lifting, invalid-index handling and recursion closure before any improved rate is claimed.
2. **Explore actual-row separation only as a constant refinement.** The slope-four Smith profile motivates a better integral construction on the restricted reciprocal rows, but the family multiplicity and recursive compatibility must be paid. It cannot by itself change the logarithmic order.
3. **Optimize depth-dependent schedules only after preserving finite guards.** The two-stage schedule is not claimed optimal. Extra stages may lower the coefficient, but must respect the initial degree exponent, packing slack, cumulative precision gap, and uniform leaf-depth argument. Such optimization still does not solve the primary target.
4. **Request the alternate proof sources before treating its formal claims as verified.** Check the actual theorem type against its strongest printed statement, pinned dependencies, axioms, full kernel/checker logs and machine-cost correspondence. A placeholder link is not evidence.

A Finite parameter conditions and crossover caution

The new schedule has exceptionally small packing slack:

$$1-\left(0.9+\frac{20\log_2 37}{1043}\right)=0.000106359240\ldots,$$

$$1-(0.9+6A)=\frac{31}{307685}=0.000100752392\ldots.$$

Thus factors such as $s^{-0.00010075}\,\mathrm{polylog}(s)$ tend to zero extremely slowly. The proof is an eventual asymptotic certificate, not an optimized finite threshold. No numerical global crossover s_0 is claimed.

For a finite implementation, test the actual integer parameters: all retained taper levels must be positive; their count must fit the packet exponent; the matching potential must start below one; padding must satisfy the desired child-size guard; the new degree $2L$ must fit the interface; the field extension must dominate deformation degree; and every precision and entry-cost bound must hold. Direct summation handles exceptional sizes or failed guards. The proof shows these guards eventually hold uniformly, so fallback is confined to a finite exceptional range for fixed interface constants. This is an existence/complexity statement, not a supplied finite solver implementation.

The additive gap $g=(\lceil\log_2\lambda\rceil)^2$ is large enough to make $\lambda^B 2^{-g}=o(1)$ for every fixed B, yet its cumulative cost $O((\log\lambda)^3)$ is $o(\lambda)$. This separation is central: one must not replace the gap by a fixed constant without a new inherited-precision proof.

B Selected source anchors and archive contents

All canonical research references use the frozen project under

`Computability/SubexponentialSAT/Research/`.

The archive includes all six new report TeX files, the four unchanged OpenAI quantitative/source TeX files, the alternate manuscript PDF, the upstream license, exact provenance maps, the new proof notes and checkers, and this report's editable source and PDF. The seven original ZIP files, the duplicate OpenAI PDF, full repository path TSV and temporary rendering images are intentionally omitted. Their identities remain in the authenticated intake and relocation ledgers.

OpenAI quantitative contraction anchors are: arithmetic and canonical creation precision (lines 11–72); packing and packet arrangement (114–171); deformation/integrality (183–241); Hermite identity and valuation loss (274–323); groups and column tiles (385–451); child factorization and entry descriptions (485–554); time/space induction (558–619). The quantitative reduction interface explicitly permits bipartite multigraphs. Line numbers refer to the bundled immutable source, not a moving branch.

For the supplied reports, useful labels are:

- Contraction: `lem:taper`, `lem:precision`, `thm:packet`, `thm:main`.
- Cauchy: `lem:identity`, `lem:factorial`, `lem:actual-domain`, `thm:weights`, `prop:recursion`, `lem:entropy`.
- Precision: `thm:precision`, `thm:optimal`, `hyp:acceleration`.
- Parity: `ass:LF`, `lem:extension`, `thm:conditional`.
- ETH density: `eq:smith` and the limits/research sections.

The exact Smith profile is credited to ETH density; the present addition is its embedding in actual packet slots and the explicit modular approximation-rank consequence.

References

[1] ProveIt, Canonical SubexponentialSAT project, frozen 27 October 2026.

[2] OpenAI, *The Exponential Time Hypothesis is False*, revised 25 October 2026. Pinned vendored source and provenance.

[3] Factorial Cauchy mode improvement packet.

[4] Quantitative contraction improvement packet.

[5] SAT density improvement packet; ETH density refinement packet.

[6] Precision refinement packet.

[7] Scalar parity refinement packet.

[8] Prior asymptotic analysis packet, already included among the arrivals.

[9] *A* $2^{o(n)}$ *Time Algorithm for 3-SAT*, attributed in the PDF to Claude / Anthropic, 26 October 2026. Pinned alternate manuscript.

[10] Isabelle Archive of Formal Proofs, Expander Graphs MGG, inspected during the source audit on 27 October 2026.